Math 70
Week 7

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May 8, 2022

1 Problems

Example 1.1. Let $X \sim Poiss(\lambda)$. Compute $Var(X)$.

Example 1.2. (Old MT Problem) Given a constant c, suppose X is a continuous RV with pdf

$$f(x) = c(x+3), \qquad -2 < x < 1$$

Find c.

Example 1.3. (5.1.3 in PSI) Let X have the density $f(x) = 4x^3,\ 0 < x < 1$. Find the density of $Y = x^2$. (A)1 (B)$2y$ (C)$3y^2$ (D)$4y^3$ (E)$5y^4$

Example 1.4. (5.1.7 in PSI) Assume $\theta \in \mathbb{R}^+$. The density of X is $f_X(x) = \theta x^{\theta-1}$ for $0 < x < 1$. Let $Y = -2\theta \ln X$. How Y is distributed?

(A)6 (B)36 (C)12 (D)14.7 (E)$\frac{144}{7}$

Example 1.5. (Old MT problem) Let X be a Binomial random variable with $E[X] = 30$ and $Var(X) = 21$. Calculate $F(4) - F(3)$, where F is the CDF of X.